

Central Washington University

Risk Assessment

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Executive Summary

This report covers Central Washington University (CWU) and their risk assessment. It starts by introducing CWU with their mission statement and some of their ideals and goals. CWU's services center around accommodating for and educating its students. The reason for the risk assessment is due to rising cyber-attacks targeting universities. CWU's crucial systems include the Student Information System (SIS), Financial and Payroll, Identity and Access Management, and Network Infrastructure. Students, faculty, staff, and others are all dependent on these systems in their everyday lives. Tables including threats and descriptions are provided for administrative, technical, and physical threats. RIIOT was used to gather this information and a description of how is provided. Relative risk analysis is calculated for many of the previous risks and is used to create a heat map for visualization and analysis. Recommended controls and safeguards are provided for each threat included in the heat map. Final recommendations are made about the importance of policy and keeping it up to date.

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Company Introduction

Mission Statement

Central Washington University (CWU) has a mission statement that reads “To build a community of access and opportunity, Central Washington University fosters high impact practices, sustainability, and authentic community partnerships that are grounded in meaningful relationships.” (Central Washington University, 2025). CWU is a public university that was founded in 1891 with their main campus located in Ellensburg, Washington. It has a diverse student population and tries to create an inviting environment where students feel like they belong (Central Washington University, 2016). CWU offers a wide range of undergraduate and graduate programs including online and hybrid options.

Services

Central Washington University’s main services center around the needs of their students and faculty. Academic advising is offered for students to help them plan their future at the school and organize their class schedule. CWU has an extensive catalog of classes offered for students to educate them on the subject of their choosing. Housing services are offered for students and is required for freshman. This includes dormitories and on-campus apartments at various price levels. Other services that CWU provide include dining-halls, a health center, recreation areas, internet, and on-campus laptop rental and desktops. Because CWU is an education center, they handle a multitude of sensitive information from finances and financial information to medical records. Because of this, it is crucial that CWU is diligent in protecting their student and faculty’s information.

Reason for Risk Assessment

Central Washington University is at an increasing risk of cyberattacks, along with other higher education centers. In 2021, higher education and research sectors were the top target for cyberattacks. Compared to 2020, in 2021 there were 75% more cyberattacks. That is 25% higher than the industry average. Because universities often don't have the same amount of resources dedicated to cybersecurity as businesses can afford, they can be an easy target. Public universities are often alluring because of their potential to store sensitive government data (Miller, 2022). Because education centers revolve around sharing knowledge, they can be easier to breach as well. 74% of attempted cyber-attacks against universities are successful compared to 68% in business, 61% in healthcare, and 57% in finance (Miller, 2022). This means that CWU is a prime target for a cyber-attack and could make good use of a Risk Assessment.

Critical Systems

Because Central Washington University has such a wide range of services and technologies, they have many critical systems. If many of these systems were to be compromised, it would spell catastrophe for CWU.

Priority and Risk

FIPS 199, Standards for Security Categorization of Federal Information and Information Systems is the reference used to sort CWU's following critical systems (Evans et al., 2004). Central Washington University has four core critical systems. The following list ranks these systems based on their FIPS 199 risk level.

1. High Risk: Student Information System (SIS)

The Student Information System is critical for Central Washington University's operations. It contains sensitive personal information about students including Social Security Numbers, contact information, financial aid information, academic records, and contact information of both students and their families (Ellucian, 2024). The disruption of the SIS could stop course enrollment, advising, and even graduation.

2. High Risk: Finance and Payroll

Finance and Payroll handle very sensitive information like bank details, payroll, Social Security Numbers, personal information, and tax information (Oyster, 2020). This is all information that would be devastating to leak. If this system got taken down, it would mean employees wouldn't be paid, tuition and revenue couldn't be calculated, any vendor services would halt, and financial aid would stop.

3. High Risk: Identity and Access Management (IAM)

Identity and Access Management systems are high risk because of the wide range of information it stores and the large role it plays in all of the university's systems. IAM is the cornerstone of all authentication in the university so its outage would mean that almost all other systems would go out. Because IAM is used in almost every university system, it makes it a large target with many different ways to get to it. Because IAM has such a large turnover of users, it makes misconfigurations and successful social engineering more likely (CyberArk, 2025).

4. High Risk: Network Infrastructure

The Network Infrastructure is crucial for almost all of the university's operations. Without internet connection, none of the systems would be able to function, shutting down the university. The likelihood of the network being targeted is extremely high because it is a single point of attack that would cripple the whole organization. There are also many levels of the network that can be targeted and a multitude of threats that can easily take it down. Because the network is so complex, it is hard to monitor suspicious activity and also hard to patch and locate vulnerabilities.

These critical systems were ranked based on FIPS 199 standards. These standards evaluate a system's risk based on how much it effects Confidentiality, Integrity, and Accessibility (CIA) (Evans et al., 2004).

System Dependencies

There are many people who depend on these systems in their daily school and work routines. The following is each critical system along with an evaluation of who depends on it and how an outage might impact them.

Student Information System:

- Students depend on the SIS for their grade reports, financial aid, course registration, and schedules.
- Faculty need the SIS for their class rosters, and grading.
- The financial aid office needs the SIS to evaluate aid packages, student aid eligibility, and aid reporting.

- Others who depend on the SIS include advisors, admissions, the registrar's office, and more.

Finance and Payroll:

- The finance office depends on this system in order to calculate their budget, complete audits, and accounting.
- Payroll needs this system to pay the university's employees, file tax documents, and issue benefits.
- Faculty need the payroll system so they can get paid and access their employment records.

Identity and Access Management:

Everyone uses IAM because it touches every system in the university. Students, faculty, staff, and even contractors would not be able to access anything in their systems if IAM were to go down. This would essentially stop the university from functioning all together.

Network Infrastructure:

Just like Identity and Access Management, everyone in the university relies on the network infrastructure. The network being compromised would cause an outage of all systems. Even worse, things like security cameras and alarms meant to keep people safe would stop functioning.

Administrative Threats

The following table illustrates some of the administrative threats that Central Washington University could face.

Table 1: Administrative Threats

<i>Threat</i>	Description	Potential Impact
<i>Decentralized IT Management</i>	The IT management of each department is independently secured	Gaps and inconsistencies across the university's cybersecurity methods.
<i>Outdated Policies</i>	The university's policies haven't been refreshed and reflect old standards and needs.	Not complying with FERPA and having inconsistencies in their system's cybersecurity methods.
<i>Poor Security Training/Awareness</i>	Staff, faculty, and students are not aware of or taught about cybersecurity threats and phishing.	Sensitive information exposed, unauthorized access to a system granted, and malware infection.

Technical Threats

The following table illustrates some of the technical threats that Central Washington University could face.

Table 2: Technical Threats

Threat	Description	Potential Impact
<i>Malware, Viruses, and Trojans</i>	Software made to harm a system and gain unauthorized access.	Data being damaged, stolen, and leaked. Spread of the virus from initial device to others. Device being unusable.
<i>Social Engineering</i>	Emails and communications designed to deceive a person into offering credentials and sensitive data.	Unauthorized access to a system, credentials being stolen, malware/ransomware infection.
<i>Denial of Service (DoS)</i>	An attack designed to overwhelm a network with more connections than it can handle	Services being unavailable like email, and campus systems.

Physical Threats

The following table illustrates some of the physical threats that Central Washington University could face.

Table 3: Physical Threats

Threat	Description	Potential Impact
<i>Unauthorized Physical</i>	An unauthorized person using	Stolen information/devices,

<i>Access</i>	a method like piggybacking/tailgating to enter a restricted area.	shoulder surfing, tampering with servers and devices.
<i>Theft</i>	An equipment or device being stolen from a public or private area.	Costly replacements, stolen sensitive information, disrupting instruction and learning.
<i>Power/Utility Failures</i>	Power, networks, and servers going out.	Safety concerns that come with power outages as well as the interruption of classes, research, and administrative system functions.

Gathering Data

The RIIOT methodology assisted in structuring and executing this research. RIIOT stands for Review, Interview, Inspect, Observe, and Test. The following is a description of each how each was implemented (Landoll, 2022).

Review – I reviewed security different security threats as well as what critical systems could look like. I refreshed myself on CWU’s standards and policies as well as their past security incidents.

Interview – I talked to IT personnel and professors about what kinds of security risks CWU is likely to face.

Inspect – I looked at how in depth CWU’s security controls are and what might fall through the cracks.

Observe – I observed how often there were network issues or canvas outages.

Test – I tested the student systems for security and efficiency.

Risk Analysis

OCTAVE Allegro can be used to calculate relative risk scores. To do this, the actor, means, motive, outcome, security requirements, probability, consequences, and severity are all considered and used to calculate a final relative risk (Alberts et al., 2007)

For example, we calculated the relative risk score of having Decentralized IT Management:

Likelihood: Medium (2)

Operations: 4

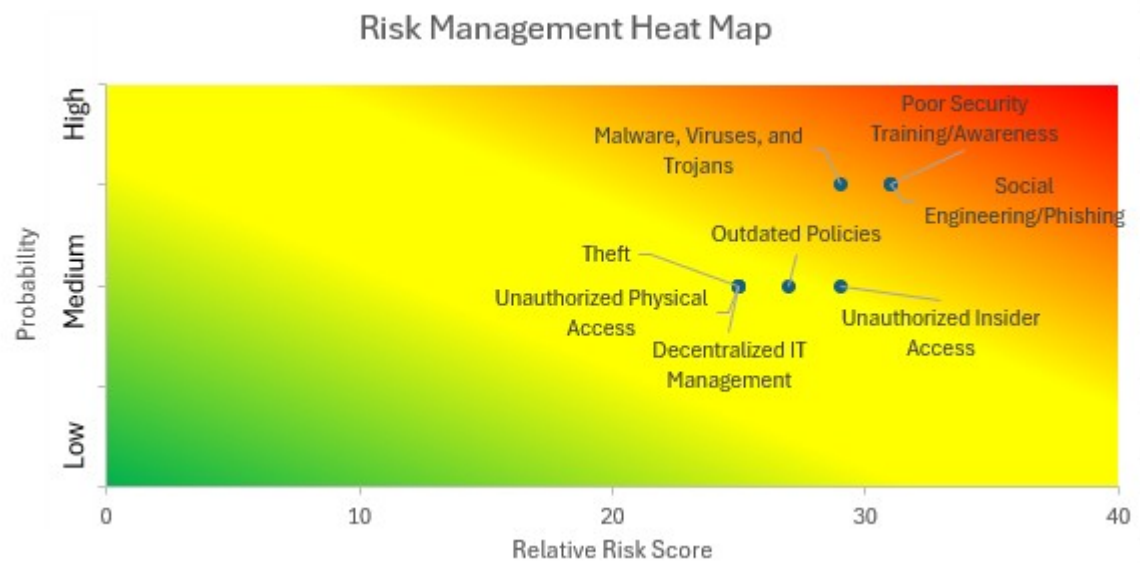
Reputation: 3

Confidentiality: 4

This resulted in a high relative risk score of (25/35).

Relative risk scores based off of the previous threat tables that were made were put into a heat map to help visualize their risks and probabilities.

Table 4: Risk Management Heat Map



Analyzing the heat map, we can see how the relative risk scores and probability affect the severity of an issue. The green section of the graph represents threats with low probability and low relative risk. These threats are rare, and when they do happen, not very problematic. The yellow section represents threats of a moderate relative risk and a medium probability. The red section represents threats with high probability and high relative risk. These are the ones to watch out for the most. We can see that the threats in this graph with the highest probability and relative risk score are “Poor Security Training/Awareness” and “Social Engineering/Phishing”.

Controls and Safeguards

Using our calculation and analysis of relative risk scores, we came up with some controls and safeguards. We used NIST SP 800-53 to help us determine what the best controls and safeguards were for these threats.

- ♣ Poor Security Training/Awareness

- o Control – NIST AT-2
- o Action - Mitigate
- o Train students, staff, and faculty to have better cyber security habits and watch for phishing attempts.

♣ Social Engineering

- o Control – NIST AT-2
- o Action - Mitigate
- o Train students, staff, and faculty to be wary of phishing and know what to look out for.

♣ Malware, Viruses, and Trojans

- o Control – NIST SI-3
- o Action - Mitigate
- o Implement anti-virus, anti-malware, intrusion detection, and firewalls to keep computers safe.

♣ Outdated Policies

- o Control – NIST PL-2
- o Action - Mitigate
- o Review and edit policies regularly so that they support standards.

♣ Unauthorized Insider Access

- o Control – NIST AC-6
- o Action - Mitigate
- o Limit access as much as possible and monitor who is accessing sensitive systems and information.

♣ Decentralized IT Management

- o Control – NIST CM-2
- o Action - Mitigate
- o Create guidelines for IT management and monitor each department to ensure consistency across the board.

♣ Unauthorized Physical Access

- o Control – NIST PE-2
- o Action - Mitigate
- o Require some kind of monitored entry that requires each employee to verify their identity before entering.

♣ Theft

- o Control – NIST PE-5
- o Action - Mitigate
- o Physically secure/monitor devices and make sure all important information stored on them is encrypted.

♣ Denial-of-Service

- o Control – NIST SC-5
- o Action - Mitigate
- o DoS detection systems and firewalls as well as a good incident response plan.

♣ Power/Utility Failures

- o Control – NIST CP-10
- o Action - Mitigate

- o Instill solid disaster recovery plans, have a backup power source, and have alternate processing sites.

Recommendations

It is clear that CWU is currently in a vulnerable position. It is important that they are prepared for possible cybersecurity threats like these. The following is a recommendation that might help with CWU's preparedness and security.

Policy Improvement:

It is clear that CWU needs some policy improvements. When reviewing some of their security and internet use policies, many holes and inconsistencies were apparent. CWU should regularly revisit their policies and update them to their current needs. After each update it is important to thoroughly enforce these policies. There is no use in a policy that is never followed. It would also be beneficial to encourage reporting of undesirable behaviors to make sure that any new policies aren't ignored.

In conclusion, Central Washington University's greatest interest right now should be improving their cybersecurity. Because they are a university, they are a bigger target for cyber-attacks with higher success rates. They should focus on securing their critical systems, including their SIS, Finance and Payroll system, IAM, and network infrastructure.

References

Central Washington University | Vision, Mission, and Strategic Plan. (2025). Cwu.edu.

<https://www.cwu.edu/about/mission-vision/vision-mision-strategic-plan.php>

Quick Facts | Central Washington University. (2016). Cwu.edu.

<https://www.cwu.edu/about/>

Miller, E. (2022, September 20). *The State of Higher Education Cybersecurity: Top Insights and*

Trends. Ww.bitlyft.com. <https://www.bitlyft.com/resources/the-state-of-higher-education-cybersecurity-insights-trends>

Evans, D., Bond, P., & Bement, A. (2004). *FIPS PUB 199.*

<https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.199.pdf>

What is a Student Information System in Higher Ed? (2024). Ellucian.

<https://www.ellucian.com/blog/what-student-information-system-higher-ed>

What is a payroll system? How does it work? | Global HR glossary | Oyster®. (2020).

Oysterhr.com. <https://www.oysterhr.com/glossary/payroll-system>

CyberArk. (2025, August 25). *Modern identity governance and Administration (IGA).* CyberArk.

https://www.cyberark.com/what-is/modern-identity-governance-administration-iga/?utm_source=google&utm_medium=paid_search&utm_term=identity_governance_and_admin_uar_max_clicks_nam_english_us&utm_content=modern_identity_governance_administration_iga&utm_campaign=modern_iga&cq_plac=&cq_net=g&cq_plt=gp&gclid=Cj0KCQiA9OnJBhD-ARIsAPV51xNJt7s3T-5Db1xciZawwKgjiyaxso_D-

u8hIJQu77oQ3x7QjuhRhQaAgixEALw_wcB&gad_source=1&gad_campaignid=23326851073&gbraid=0AAAAAD_gt5FbZ9BWN_FyaHYOEfgyN1EQx

Landoll, D. J. (2022). The security risk assessment handbook: A complete guide for performing security risk assessments (3rd ed.). CRC Press. <https://Landoll, 2022.pub/the-security-risk-assessment-handbook-a-complete-guide-for-performing-security-risk-assessments-3nbsped-103204165x-9781032041650.html>

Alberts, C. et al. (2007). Introducing OCTAVE Allegro: Improving the Information Security Risk Assessment Process. Carnegie Mellon SEI. (Relative risk score definition). https://resources.sei.cmu.edu/asset_files/technicalreport/2007_005_001_14885.pdf#:~:text=relative%20risk%20score,the%20organizational%20impact%20of%20risk

National Institute of Standards and Technology (NIST). (2004). Standards for security categorization of federal information and information systems (FIPS PUB 199). U.S. Department of Commerce. <https://nvlpubs.nist.gov>

National Institute of Standards and Technology. (2020). Security and privacy controls for information systems and organizations (SP 800-53 Rev. 5). U.S. Department of Commerce. <https://csrc.nist.gov/publications/detail/sp/800-53/rev-5/final>